

Efficiency Corrections Comparison: Current vs. Previous Results

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March 27, 2026

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1 Methodology: Efficiency and Uncertainty Calculations

This section details the updated statistical methodologies used to evaluate the average efficiencies and propagate their associated uncertainties within each kinematic mass and x_F bin.

1.1 Reconstruction Efficiency and Correlated Uncertainties

The reconstruction efficiency $E_{\text{reco},i}$ for a single event i is interpolated from a global tracking efficiency curve evaluated at the chamber occupancy parameter, D_1 . Because multiple events within the same kinematic bin frequently fall between the same D_1 occupancy, their interpolated uncertainties σ_i are mathematically correlated.

To account for this, we construct an $N \times N$ covariance matrix for the N events within a given kinematic bin:

$$\text{Cov}_{i,j} = \rho_{i,j} \sigma_i \sigma_j \quad (1)$$

where the correlation coefficient $\rho_{i,j}$ is determined by the shared occupancy:

- $\rho_{i,j} = 1$ if event i and event j share both bounding occupancy (same D_1 bin).
- $\rho_{i,j} = 1$ if event i and event j share exactly one bounding interpolation node (adjacent D_1 bins).
- $\rho_{i,j} = 0$ otherwise (assumed strictly independent).

The mean reconstruction efficiency for the kinematic bin is evaluated as the arithmetic mean of the event-by-event efficiencies:

$$\bar{E}_{\text{reco}} = \frac{1}{N} \sum_{i=1}^N E_{\text{reco},i} \quad (2)$$

By accounting for the off-diagonal covariance elements, the correctly propagated, correlated uncertainty on the mean efficiency is:

$$\sigma_{\bar{E}_{\text{reco}}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sum_{j=1}^N \text{Cov}_{i,j}} \quad (3)$$

1.2 Final Efficiency Evaluation

The mean hodoscope efficiency $\bar{E}_{\text{hodo}}$ is derived similarly. However, because its event-by-event evaluations are independent of the global tracking D_1 curve, its uncertainty is evaluated assuming strictly independent events (a purely diagonal covariance matrix):

$$\sigma_{\bar{E}_{\text{hodo}}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sigma_{\text{hodo},i}^2} \quad (4)$$

Previously, the final efficiency was calculated by multiplying $E_{\text{reco}} \times E_{\text{hodo}}$ on an event-by-event basis prior to averaging. To properly isolate and utilize the independent error characteristics of the tracking and hodoscope responses, the current methodology calculates the final efficiency as the product of the independently evaluated bin averages:

$$\bar{E}_{\text{final}} = \bar{E}_{\text{reco}} \times \bar{E}_{\text{hodo}} \quad (5)$$

The final propagated uncertainty on this product is evaluated using standard error propagation for independent variables:

$$\sigma_{\bar{E}_{\text{final}}} = \sqrt{(\bar{E}_{\text{hodo}} \cdot \sigma_{\bar{E}_{\text{reco}}})^2 + (\bar{E}_{\text{reco}} \cdot \sigma_{\bar{E}_{\text{hodo}}})^2} \quad (6)$$

2 LH2 Efficiency Corrections

2.1 Total Yield

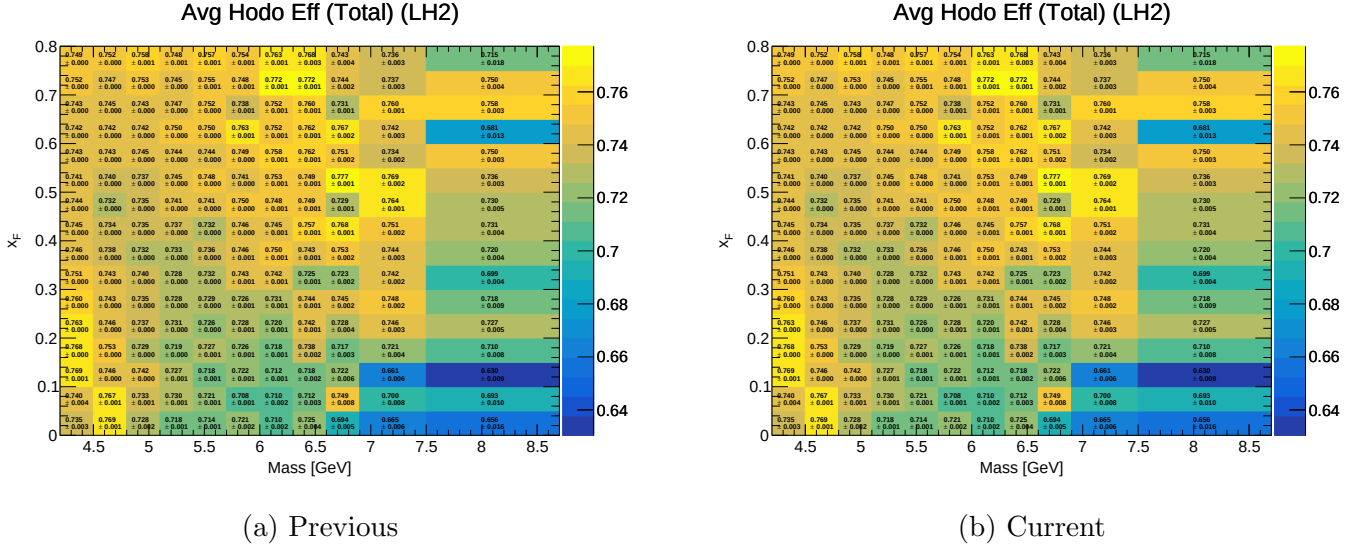


Figure 1: LH2 Total - Hodoscope Efficiency Corrections

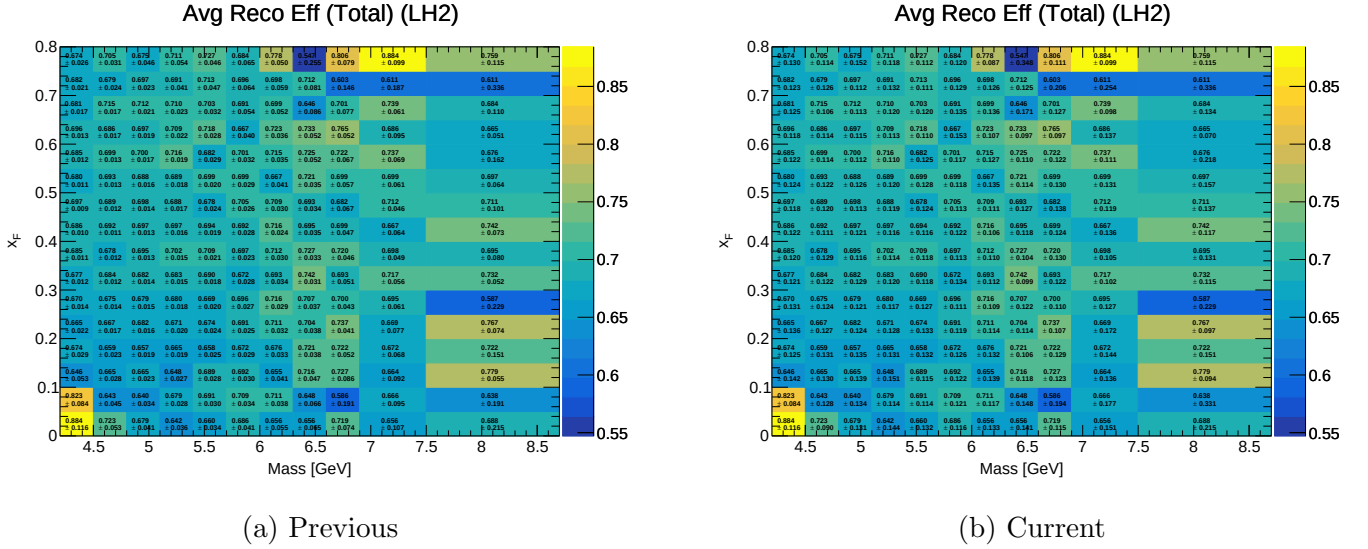
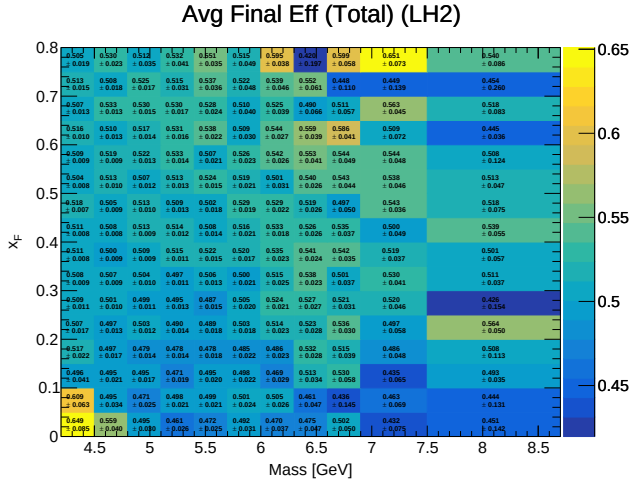
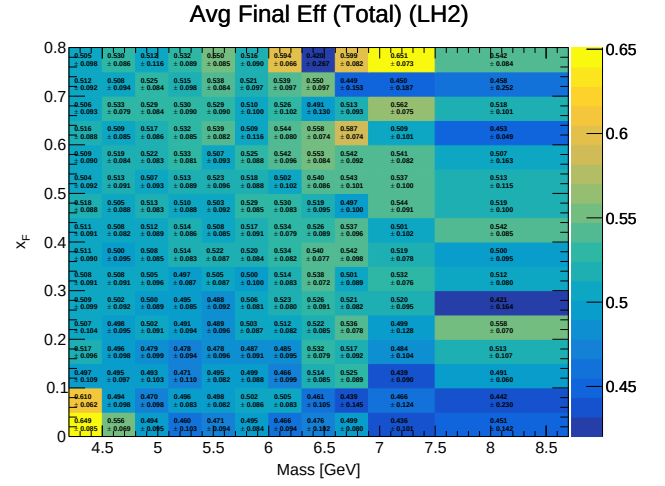


Figure 2: LH2 Total - Reconstruction Efficiency Corrections



(a) Previous



(b) Current

Figure 3: LH2 Total - Final Efficiency Corrections

2.2 Mix Yield

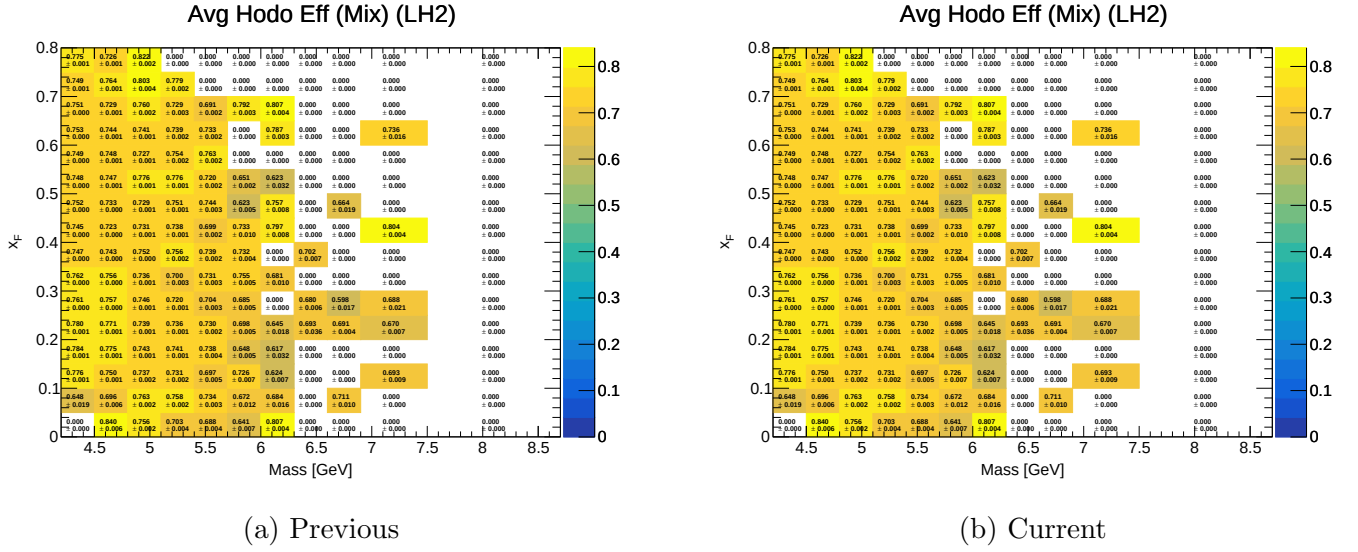


Figure 4: LH2 Mix - Hodoscope Efficiency Corrections

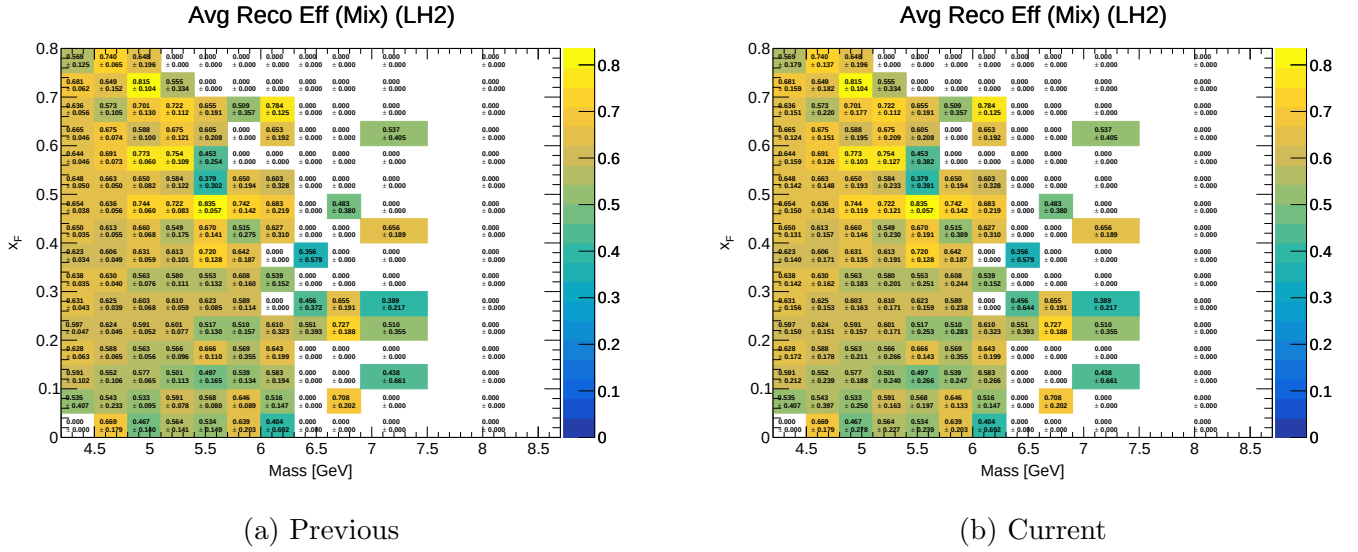


Figure 5: LH2 Mix - Reconstruction Efficiency Corrections

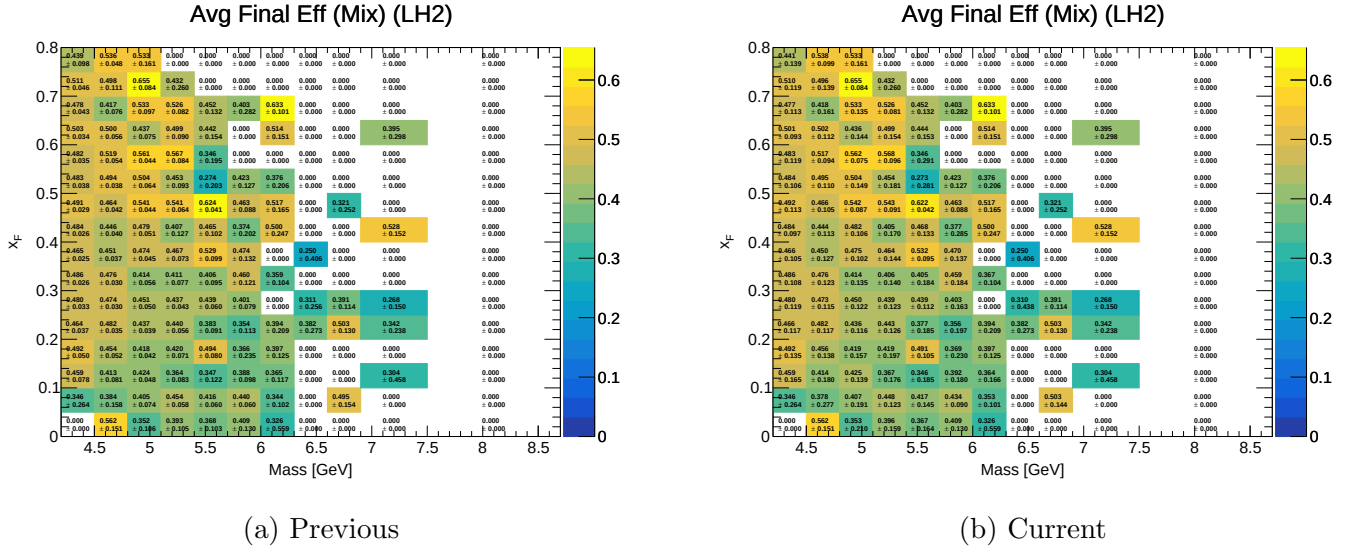


Figure 6: LH2 Mix - Final Efficiency Corrections

2.3 Signal Efficiency Corrections

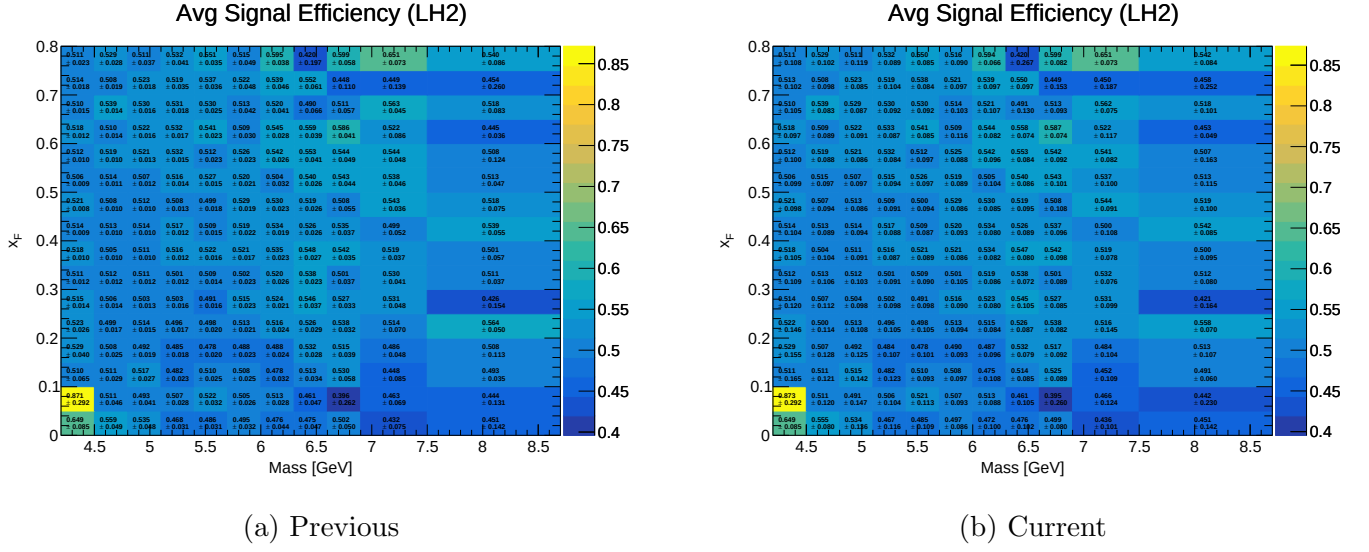


Figure 7: LH2 - Signal Efficiency Corrections

3 LD2 Efficiency Corrections

3.1 Total Yield

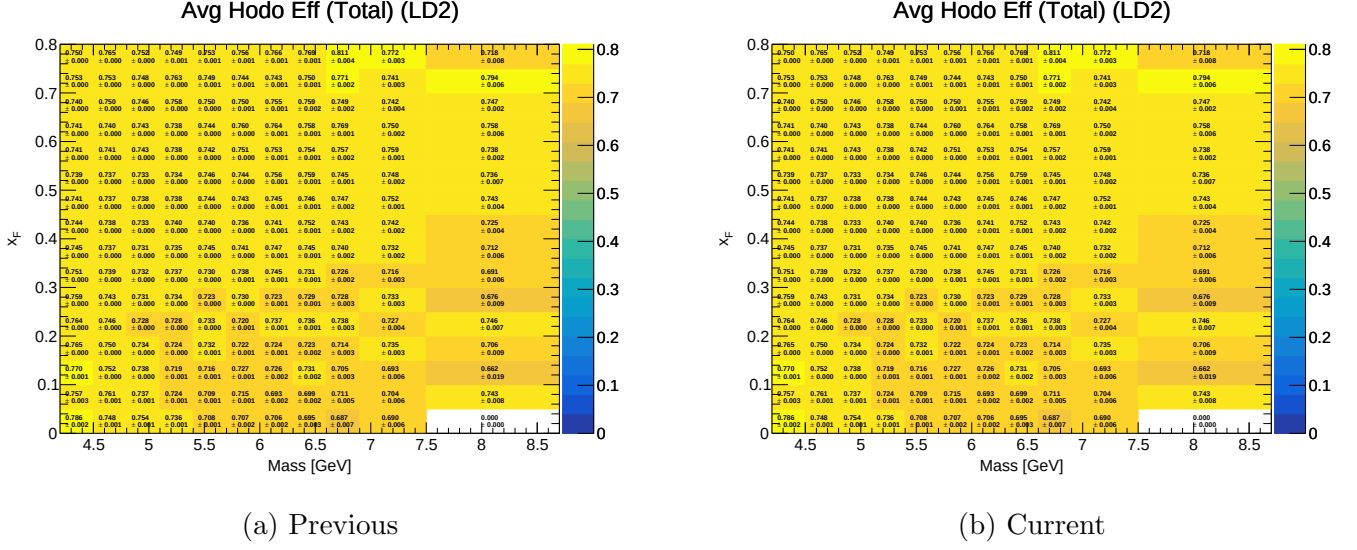


Figure 8: LD2 Total - Hodoscope Efficiency Corrections

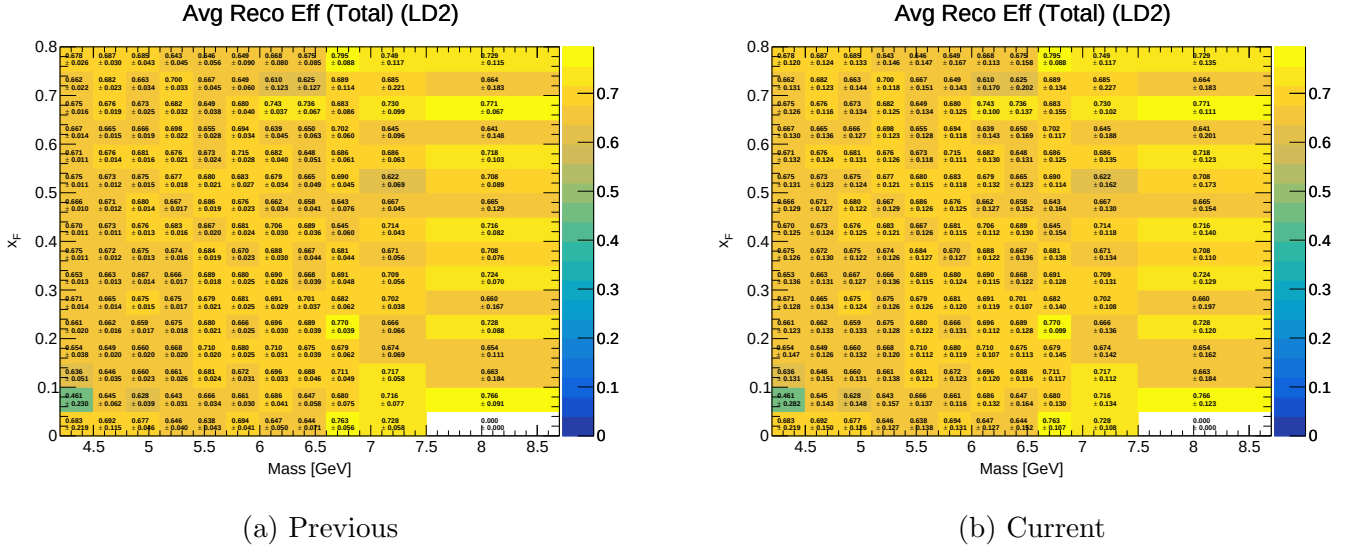


Figure 9: LD2 Total - Reconstruction Efficiency Corrections

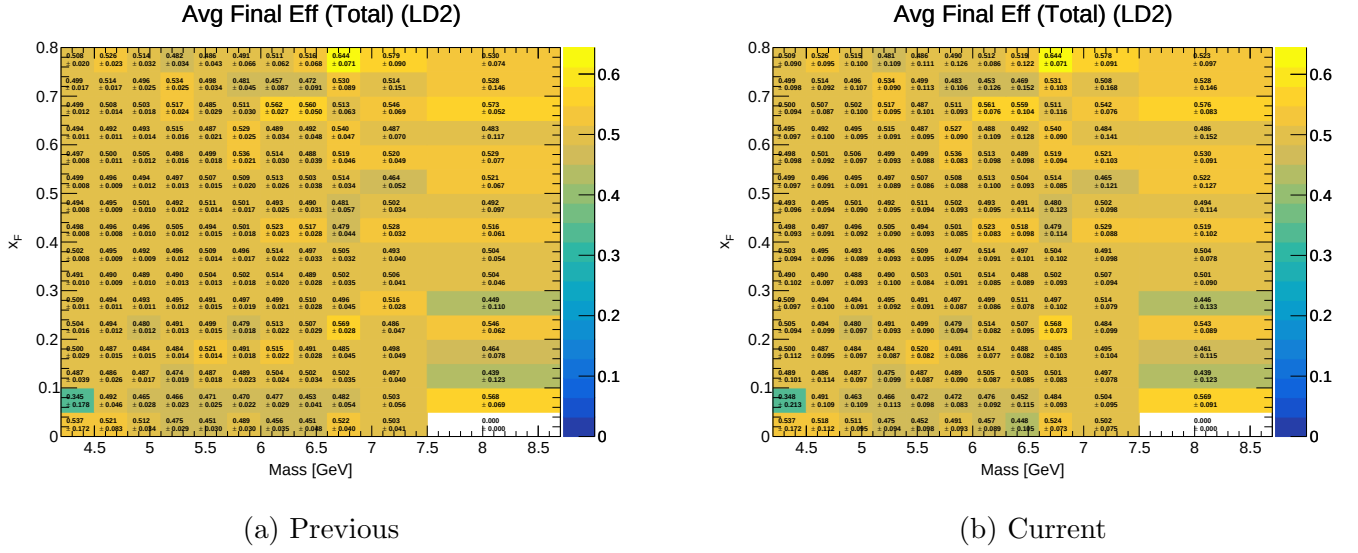


Figure 10: LD2 Total - Final Efficiency Corrections

3.2 Mix Yield

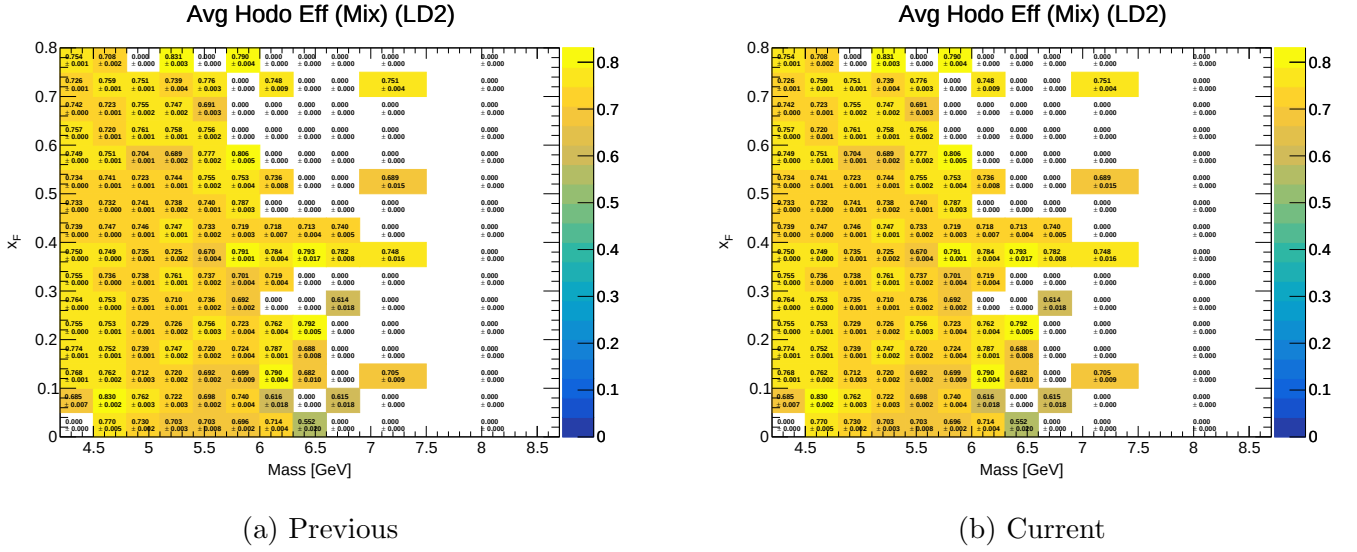


Figure 11: LD2 Mix - Hodoscope Efficiency Corrections

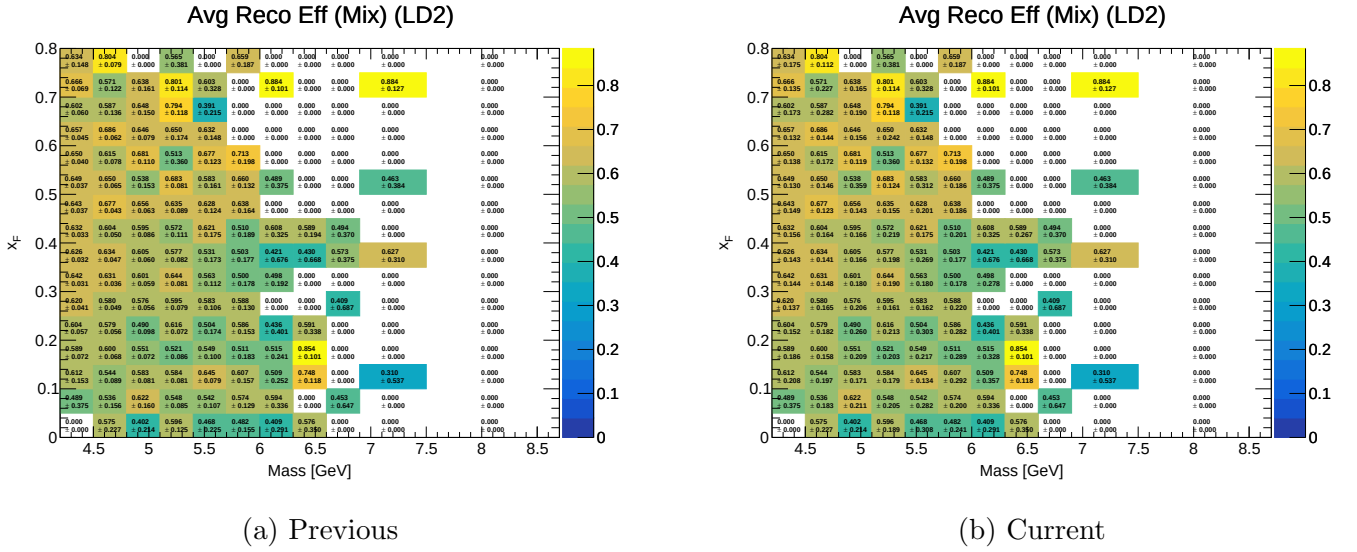


Figure 12: LD2 Mix - Reconstruction Efficiency Corrections

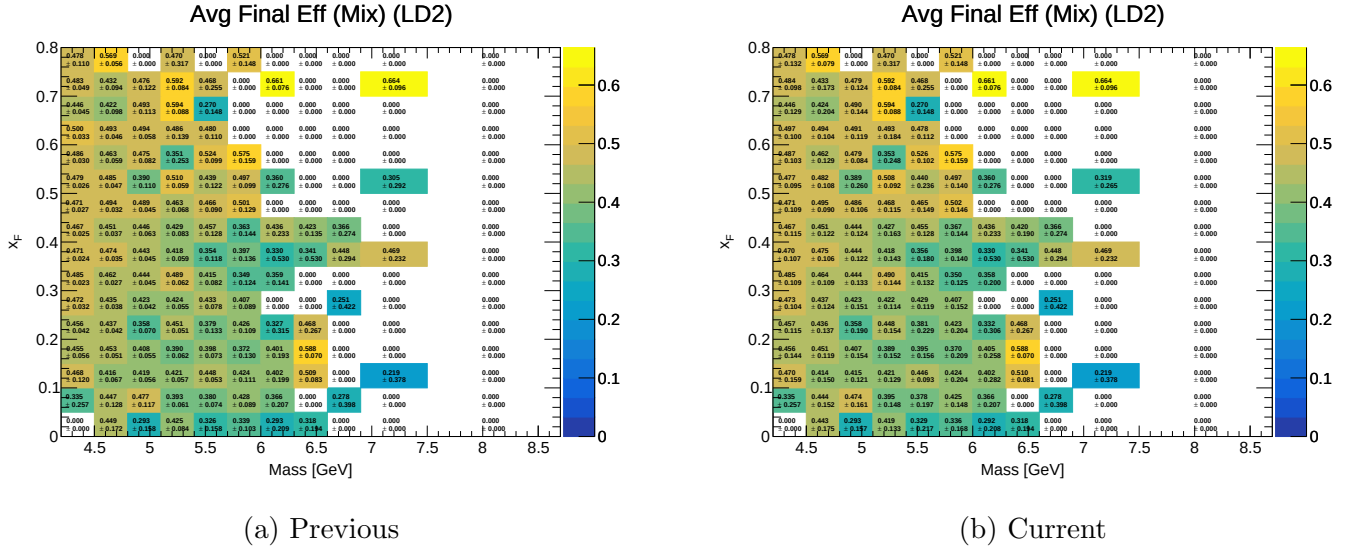


Figure 13: LD2 Mix - Final Efficiency Corrections

3.3 Signal Efficiency Corrections

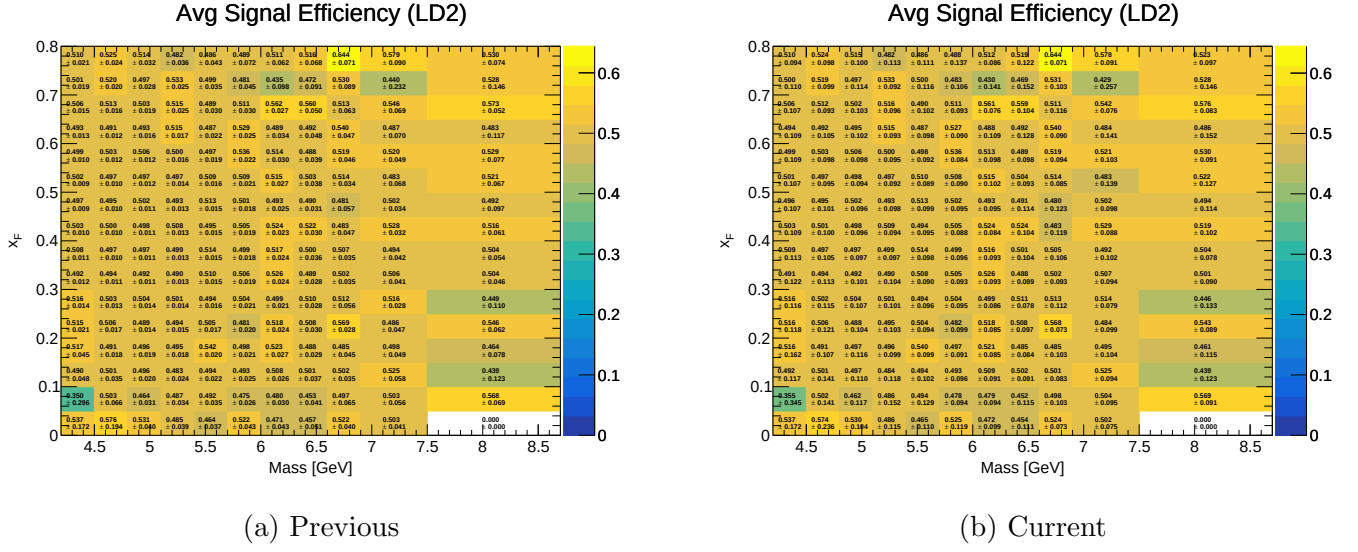


Figure 14: LD2 - Signal Efficiency Corrections

4 Flask Efficiency Corrections

4.1 Total Yield

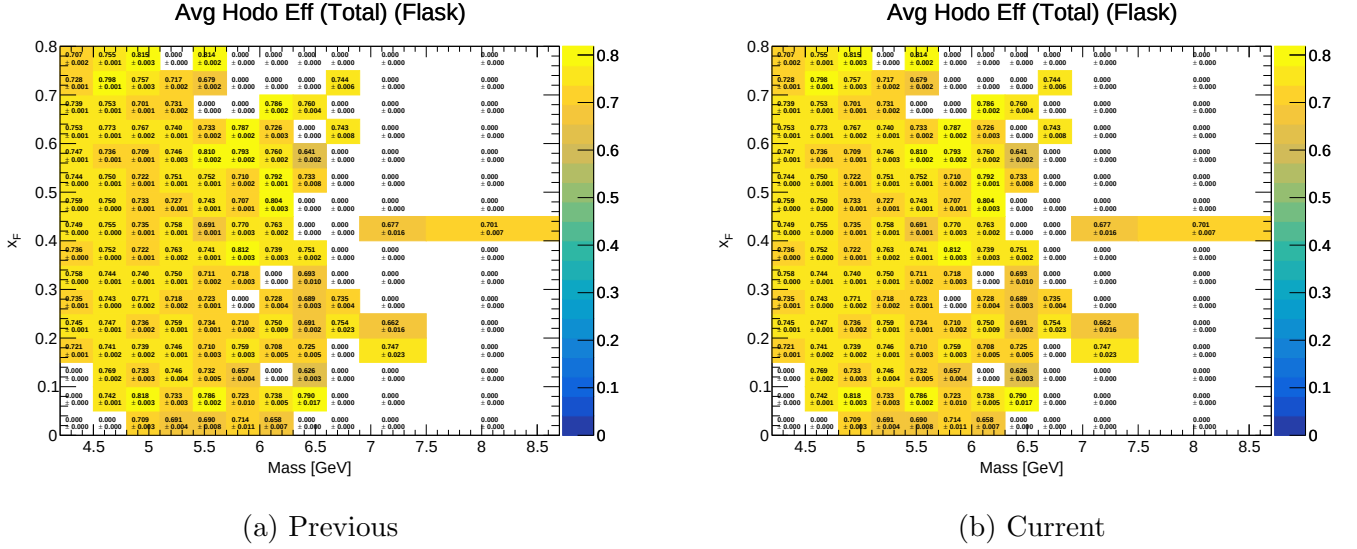


Figure 15: Flask Total - Hodoscope Efficiency Corrections

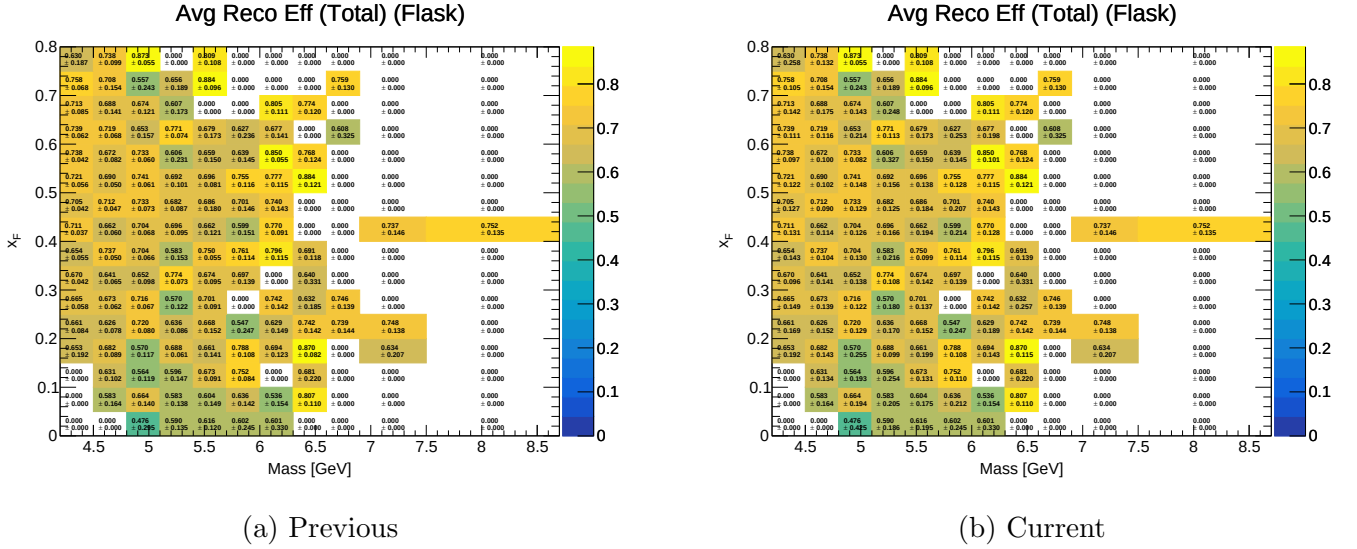


Figure 16: Flask Total - Reconstruction Efficiency Corrections

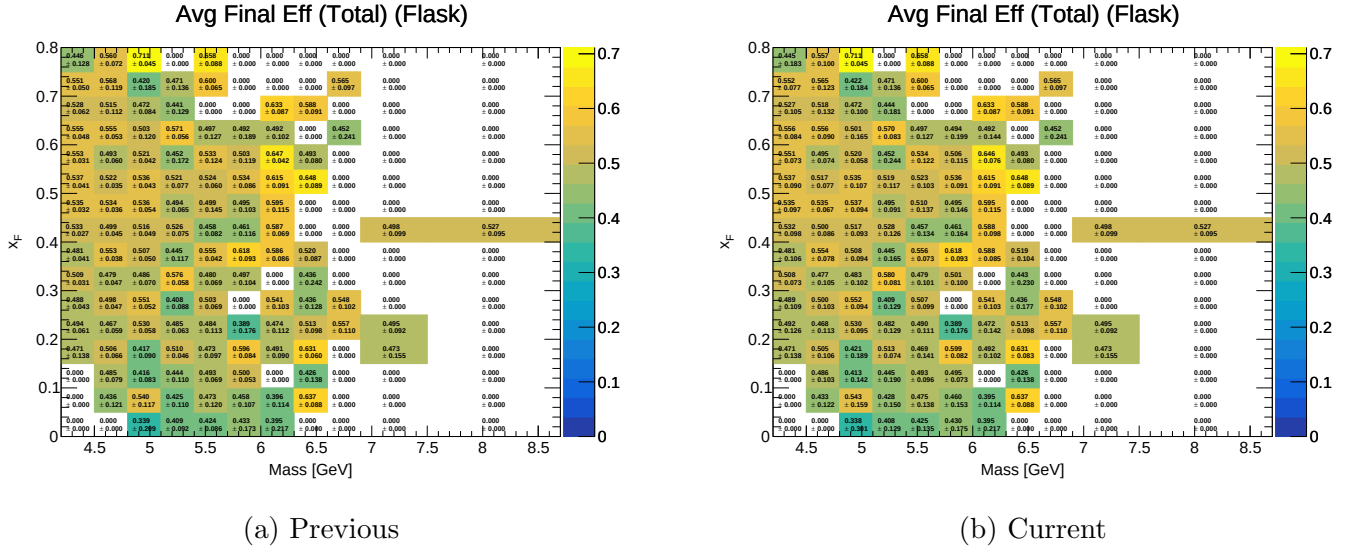


Figure 17: Flask Total - Final Efficiency Corrections

4.2 Mix Yield

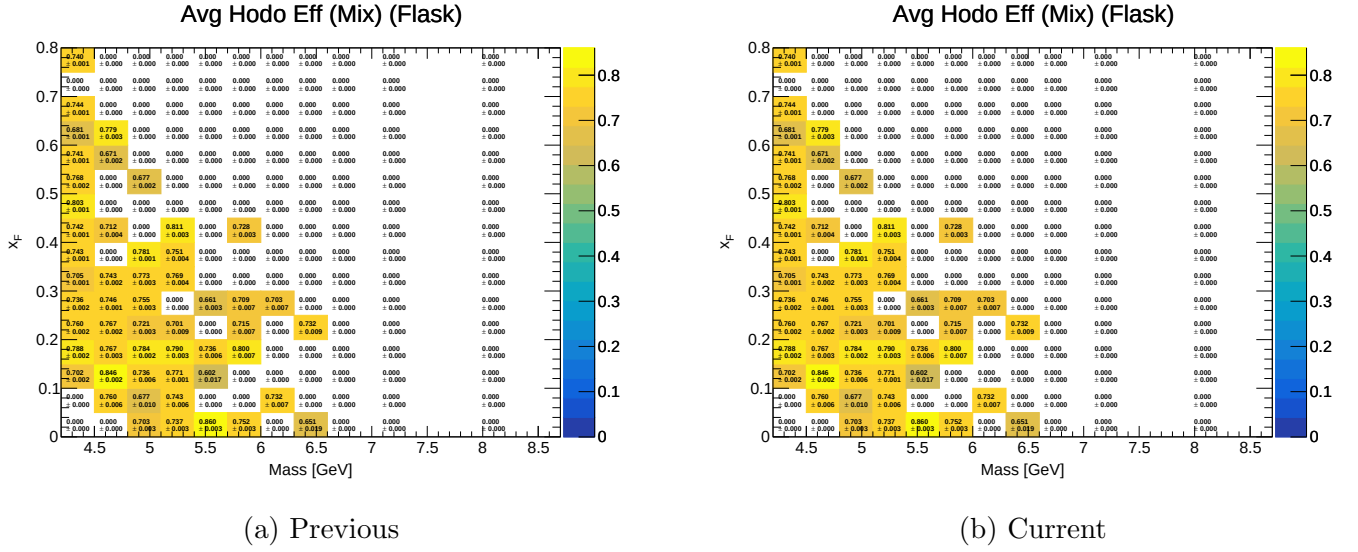


Figure 18: Flask Mix - Hodoscope Efficiency Corrections

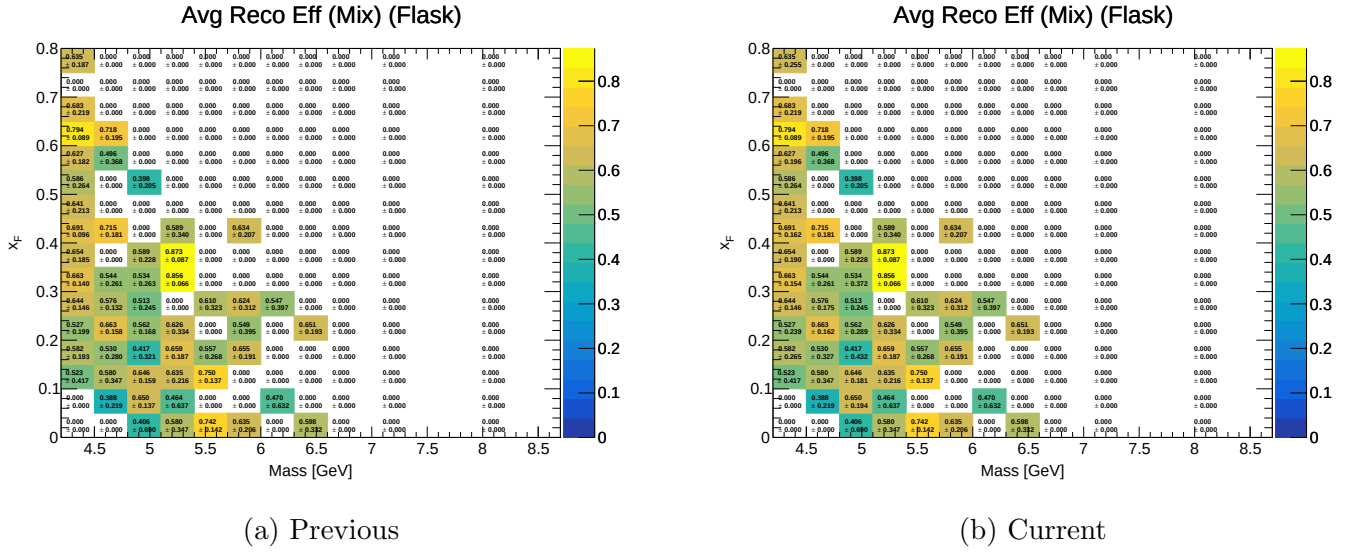


Figure 19: Flask Mix - Reconstruction Efficiency Corrections

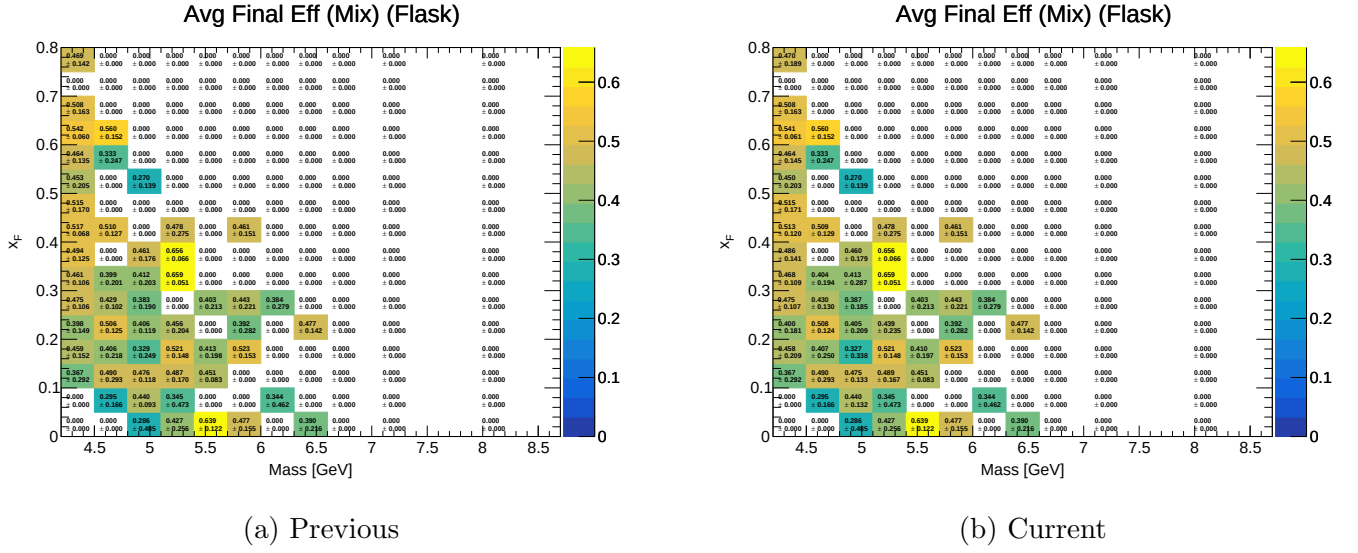


Figure 20: Flask Mix - Final Efficiency Corrections

4.3 Signal Efficiency Corrections

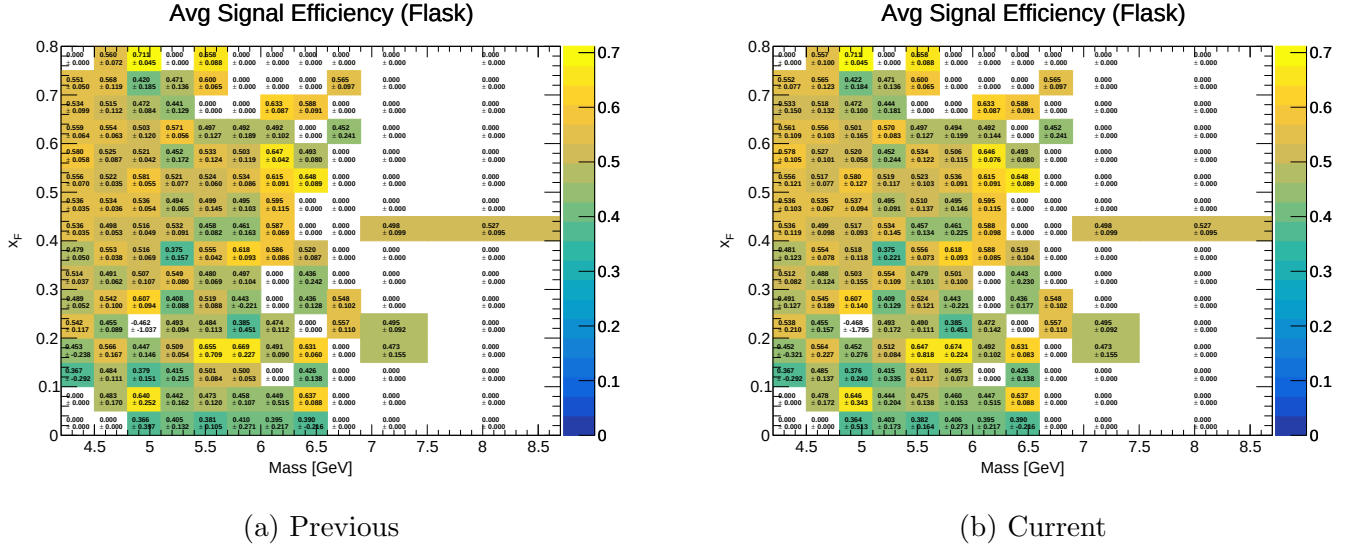


Figure 21: Flask - Signal Efficiency Corrections